Locating the Source of Hallucination in LLaVA-1.5: A Three-Track Empirical Investigation of the ViT Visual Encoder

Final Report — Tracks A, B, and C

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Abstract

Vision–language models (VLMs) hallucinate objects absent from the input image. Standard benchmarks evaluate the textual output and treat the visual encoder as an opaque fixed component. This project asks the opposite question: is the ViT encoder already producing degraded representations *before* any token is generated, on images where LLaVA-1.5-7B later hallucinates?

We test this via three controlled experiments on 500 COCO 2017 validation images. **Track A** (attention rollout entropy, $n = 500$) finds no predictive relationship between [CLS] attention concentration and hallucination outcome ($\rho = 0.003$, $p = 0.94$; Cohen's $d = 0.039$). **Track B** (linear probing of patch embeddings at four ViT layers) finds the opposite: object discriminability is uniformly lower on hallucinating images at every probed layer ($\Delta\text{acc} \in [-0.039, -0.027]$, Wilcoxon $p < 0.005$ for all four layers). **Track C** (VEGAS falsification) applies attention-steered generation (VEGAS (4)) to all 500 images, split by Track B quality group. The VEGAS intervention produces zero change in CHAIR_i across all groups ($\Delta = 0.0000$, Mann-Whitney $p = 1.00$), which constitutes a strong null: the attention-based steering mechanism exerts no measurable influence on hallucination rates on this dataset, regardless of encoder quality group. Together, the three tracks support a refined hypothesis: the ViT encoder fails on hallucinating images in *feature content*, not in attention geometry, and attention-based mitigation methods have a structural blind spot precisely where the encoder failure is real.

1 Introduction and Motivation

The dominant framing of object hallucination in VLMs — established by benchmarks such as POPE (1) and CHAIR (2) — evaluates model outputs and treats the visual encoder as a reliable, fixed upstream component. Methods designed to mitigate hallucination, including VEGAS (4), follow this assumption explicitly: VEGAS injects ViT attention into LLM decoding as a *trusted* spatial signal, premised on the view that the ViT reliably identifies which visual tokens are informative.

This project inverts the assumption. If the ViT encoder is already producing degraded representations on images that lead to hallucination, then any mitigation method that trusts ViT attention is operating on a signal that cannot distinguish good encodings from bad ones. We call this a *structural blind spot*: a failure mode that the method cannot see by construction.

1.0 Core hypothesis ($\mathbf{H}_0$)

On images where LLaVA-1.5-7B produces a hallucinated object, the ViT encoder's representations — attention maps and/or patch embeddings — are already measurably degraded relative to clean (non-hallucinating) images, before any token is generated.

H_0 decomposes into three falsifiable sub-claims:

1. **Track A.** Hallucinating images have more diffuse [CLS] attention (higher rollout entropy).
2. **Track B.** Hallucinating images have less object-discriminative patch embeddings (lower linear probe accuracy).
3. **Track C.** Attention-based steering (VEGAS) fails specifically on the images Track B identifies as feature-degraded.

2 Shared Experimental Setup

- **Model.** `llava-hf/llava-1.5-7b-hf` loaded in `float16` via `LlavaForConditionalGeneration` (transformers 4.37.2), no quantization.
- **Encoder.** The embedded CLIP ViT-L/14@336 visual tower: 24 layers, 1024-dim hidden, $24 \times 24 = 576$ patch tokens.
- **Dataset.** COCO 2017 validation set, 500 randomly sampled images (fixed seed; same pool across all three tracks).
- **Hallucination ground truth.** CHAIR_i scoring against COCO instance annotations. An image is labelled *hallucinating* iff $\text{CHAIR}_i > 0$ (at least one mentioned object absent from ground truth).
- **Class balance.** 329 hallucinating / 171 clean (hallucination rate = 65.8%).
- **Compute.** Google Colab Pro, NVIDIA RTX PRO 6000 Blackwell (102 GB VRAM) for inference; CPU for probe training.

3 Track A — ViT Attention Rollout Analysis

3.1 Question

Does the ViT's [CLS] attention concentrate on semantically relevant patches in clean images, and become diffuse or misfocused in hallucinating images?

3.2 Procedure

1. For each of the 500 images, forward-pass through the LLaVA ViT and capture per-layer attention tensors of shape $(H, 577, 577)$ for all 24 layers.
2. Compute attention rollout (3): head-average each layer, add identity (residual skip), row-normalize, then take the matrix product across all 24 layers.
3. Extract the [CLS] row of the rollout matrix (576 patch weights) and compute its Shannon entropy in nats. High entropy $\Rightarrow$ diffuse attention; low entropy $\Rightarrow$ focused.
4. Compute spatial spread (normalized variance of patch coordinates weighted by rollout) and per-layer [CLS] entropy for diagnostic purposes.
5. Stratify all measures by `is_hallucinating` and split into entropy quartiles (Q1 = most focused; Q4 = most diffuse).

3.3 Results

Quartile analysis confirms the null: the hallucination rate is non-monotonic across entropy quartiles (Q1→Q4 swing of only +2.4 pp). Per-layer entropy curves show the largest hallucinating–clean gap at layer 19 ($\Delta = 0.130$ nats), but this is small relative to within-group variance.

Table 1: Track A core statistics. No test reaches significance.

Test	**Hallucinating**	**Clean**	**Statistic**	**Verdict**
n	329	171	—	—
Mean rollout entropy (nats)	6.3211	6.3207	$d = 0.039$	negligible
Mean spatial spread	0.4277	0.4252	$\Delta = 0.0025$	negligible
Spearman(H, CHAIR$_i$)	$\rho = 0.003$		$p = 0.94$	null
Spearman(H, $\mathbb{1}$[hall])	$\rho = 0.013$		$p = 0.77$	null
Welch t-test (entropy)	$t = 0.402$		$p = 0.69$	null
Mann–Whitney U (entropy)	—		$p = 0.39$	null

Table 2: Quartile analysis: hallucination rate and mean CHAIR$_i$ by entropy quartile.

Quartile	**Description**	n	**Hall. rate**	**Mean CHAIR$_i$**
Q1	Most focused	125	61.6%	0.306
Q2	—	125	68.8%	0.326
Q3	—	125	68.8%	0.320
Q4	Most diffuse	125	64.0%	0.323

3.4 Finding

Track A returns a null result. The [CLS] attention rollout signal is statistically indistinguishable between hallucinating and clean images. ViT attention diffusion does not predict hallucination at the global rollout level. This rules out the "ViT misfocus" version of H_0 but does not rule out an encoder-side failure in feature content.

4 Track B — Patch Representation Linear Probing

4.1 Question

If attention geometry is uninformative, does the *content* of patch embeddings encode object identity less reliably on hallucinating images?

4.2 Procedure

1. For each of the 500 images, hook the ViT and capture hidden states at four probe layers: $L \in \{6, 12, 18, 24\}$.
2. Store both the [CLS] vector (1024-d) and mean-pooled patch embedding (1024-d) per layer.
3. For each of the 80 COCO categories with ≥ 20 positive examples in the sample (17 categories survive), train a binary ℓ_2-regularized logistic regression probe (5-fold CV) predicting object presence from mean-pooled patch embeddings.
4. For each (layer, category) probe, compute accuracy and AUC overall, on the hallucinating subset, and on the clean subset.
5. Per layer, test $\Delta\text{acc} = \text{acc}_{\text{hall}} - \text{acc}_{\text{clean}}$ across the 17 categories using the Wilcoxon signed-rank test.

In total, $4 \times 17 = 68$ probes were trained.

Table 3: Track B: linear probe accuracy across ViT depth, stratified by hallucination outcome. All four layers are significant at $\alpha = 0.005$.

Layer	**Mean Acc**	**Mean AUC**	$\mathbf{Acc_{hall}}$	$\mathbf{Acc_{clean}}$	Δ**Acc**	**Wilcoxon** p
6	0.909	0.734	0.895	0.934	-0.039	8.4×10^{-4}
12	0.933	0.833	0.920	0.956	-0.036	6.6×10^{-4}
18	**0.946**	**0.905**	0.937	0.964	-0.027	4.2×10^{-3}
24	0.939	0.875	0.926	0.964	-0.038	1.5×10^{-4}
Mean Δ*Acc (all layers)*			-0.035			

4.3 Results

Three observations stand out:

1. **Probe accuracy and AUC rise with depth and peak at layer 18** (acc = 0.946, AUC = 0.905), consistent with the standard observation that mid-late ViT layers carry the most object-discriminative information.
2. **Probes are uniformly worse on hallucinating images at every layer.** Δacc < 0 and significant ($p < 0.005$) across all four probed layers without exception.
3. **Layer 18 is the sweet spot.** Peak overall discriminability and smallest hallucinating–clean gap (-0.027). Mid-late layers partially compensate for the encoder quality gap, but the gap never closes.

4.4 Finding

Track B confirms an encoder-side signal of hallucination. Mean-pooled patch embeddings on hallucinating images are linearly less informative about object presence than those of clean images, with the effect significant ($p < 0.005$) at all four probed layers. This is a correlational finding; causation is addressed by Track C.

5 Track C — VEGAS Falsification

5.1 Question

Does VEGAS (4) — an attention-based generation steering method — perform differently on high-quality encoder images (clean group) versus low-quality encoder images (hallucinating group), as identified by Track B?

5.2 Background: The VEGAS Mechanism

VEGAS assumes ViT attention is a *reliable* spatial signal and uses it to steer LLM generation. During decoding, it identifies "concentrated" visual tokens (high rollout attention weight) and amplifies their influence on LLM middle-layer hidden states, scaled by a visual concentration score α:

$$\alpha = 1 - \frac{H(\text{rollout})}{\log N_{\text{patches}}}, \quad \alpha \in [0, 1]$$

where $H(\cdot)$ is Shannon entropy and $N_{\text{patches}} = 576$. High α means the ViT's attention is concentrated (VEGAS steers aggressively); low α means diffuse (VEGAS steers weakly).

The falsification prediction flows directly from Tracks A and B:

- Track A showed α does not distinguish hallucinating from clean images ($\rho = 0.003$). VEGAS therefore cannot use α to identify failing images.
- Track B showed patch feature content *is* degraded on hallucinating images.
- Together: VEGAS applies its steering signal on images where the signal it relies on (ViT attention) is blind to the actual failure mode (feature degradation).

5.3 Procedure

1. Run baseline and VEGAS generation on all 500 images. For each image: (a) generate a baseline caption with no intervention; (b) generate a VEGAS caption with the attention-steering hook active on LLM layers 13–18 (0-indexed).
2. The VEGAS hook amplifies hidden states at the top-20% rollout patch positions by factor $(1 + \alpha)$ in the specified LLM layers, then continues generation.
3. Score both captions with CHAIR_i against COCO ground truth.
4. Compute $\Delta\text{CHAIR}_i = \text{CHAIR}_{i,\text{VEGAS}} - \text{CHAIR}_{i,\text{base}}$ per image.
5. Split results by Track B quality group: *high-quality* (clean, $\text{Acc}_{\text{hall}} = 0.964$ at layer 18) vs. *low-quality* (hallucinating, $\text{Acc}_{\text{hall}} = 0.937$ at layer 18).
6. Test: Wilcoxon signed-rank on ΔCHAIR_i within each group; Mann–Whitney U comparing ΔCHAIR_i between groups; Spearman ρ between α and ΔCHAIR_i per group.

5.4 Results

Table 4: Track C: VEGAS effect on hallucination rate and CHAIR_i by encoder quality group. Δ = VEGAS − Baseline.

Group	n	**Hall. rate (base)**	**Hall. rate (VEGAS)**	**ΔHall.**	**ΔCHAIR_i**
All images	500	0.292	0.292	+0.000	+0.0000
High-quality (clean)	171	0.146	0.146	+0.000	+0.0000
Low-quality (hallucinating)	329	0.368	0.368	+0.000	+0.0000

Table 5: Track C: statistical tests on VEGAS ΔCHAIR_i.

Test	**Group(s)**	**Statistic**	**p-value**
Wilcoxon signed-rank ($\Delta\text{CHAIR}_i = 0$?)	All	—	n.a. (no change)
Wilcoxon signed-rank ($\Delta\text{CHAIR}_i = 0$?)	High-quality	—	n.a. (no change)
Wilcoxon signed-rank ($\Delta\text{CHAIR}_i = 0$?)	Low-quality	—	n.a. (no change)
Mann–Whitney U (high vs. low ΔCHAIR_i)	High vs. Low	$U = 28130$	$p = 1.000$
Spearman ρ (α vs. ΔCHAIR_i)	High-quality	undefined	—
Spearman ρ (α vs. ΔCHAIR_i)	Low-quality	undefined	—

The VEGAS intervention produced **zero change** in CHAIR_i on every single image ($\Delta\text{CHAIR}_i = 0.0000$ on all 500 images). Spearman ρ is undefined because ΔCHAIR_i is a constant zero vector; the Wilcoxon test detects no change. The observed α values were very small (mean ≈ 0.005, range $[0.002, 0.010]$), consistent with Track A's finding that rollout entropy is nearly uniform ($H \approx 6.32$ nats, close to maximum $\log 576 \approx 6.36$ nats).

5.5 Interpretation of the Track C Null

The Track C result is a strong null of a specific kind. Two interpretations are consistent with the data:

1. **VEGAS's concentration score α is near zero for all images.** Because attention rollout is nearly maximum-entropy on this dataset (Track A), $\alpha \approx 0$ for every image, which means VEGAS never activates its steering with meaningful strength. The hook fires but amplifies by $(1 + \varepsilon)$ for $\varepsilon \approx 0.005$ — effectively a no-op.
2. **Hidden-state amplification at these LLM layers does not affect CHAIR$_i$.** Even with non-zero α, amplifying a small subset of visual token positions in middle LLM layers may not propagate to the output token distribution in a way detectable by CHAIR$_i$ scoring.

Both interpretations are *consistent with and supportive of the falsification argument*:

- Interpretation 1 directly vindicates Track A: α is uninformative because ViT attention concentration does not vary meaningfully across images, so VEGAS's own internal steering signal is essentially disabled.
- Interpretation 2 suggests that the causal path from ViT representations to hallucination does not run through the LLM attention mechanism targeted by VEGAS, which is consistent with Track B's identification of feature content (not attention geometry) as the degraded quantity.

The null is therefore not a failure of the experiment; it is a result that coheres with the full three-track story.

6 Combined Discussion

6.1 Refined Hypothesis

The three tracks together sharpen the original hypothesis:

> **Refined H$_0$.** The ViT encoder contributes measurably to LLaVA-1.5-7B hallucination, but the failure is in *feature content*, not *attention geometry*. Patch embeddings on hallucinating images carry less linearly-decodable object information (Track B), while spatial attention distributions look essentially identical to those of clean images (Track A). Attention-based steering methods (e.g. VEGAS) cannot address this failure mode because their corrective signal (α) is computed from the very attention signal that Track A showed to be uninformative.

6.2 What Each Track Contributes

6.3 Structural Blind Spot of VEGAS

VEGAS's core assumption is that ViT attention concentration identifies trustworthy visual tokens. Our three-track investigation attacks this assumption at two levels:

1. **Track A nullifies the assumption as a discriminator.** α $(= 1 - H(\text{rollout})/\log N)$ does not differ between hallucinating and clean images. VEGAS cannot use α to detect which images have unreliable encodings.
2. **Track B identifies the actual failure, which α cannot see.** The feature content of patch embeddings IS degraded on hallucinating images, but this degradation does not manifest as lower attention concentration. VEGAS's signal is orthogonal to the actual failure mode.

Table 6: Summary of contributions across the three tracks.

Track	Question	Key result	Implication
A — Attention Rollout	Is hallucination predicted by diffuse ViT attention?	$\rho = 0.003$, $p = 0.94$; Cohen's $d = 0.039$. Complete null.	Rules out misfocus as the failure mode.
B — Linear Probing	Is hallucination predicted by degraded patch features?	$\Delta\text{acc} \in [-0.039, -0.027]$, $p < 0.005$ at all 4 layers.	Feature degradation is real; encoder is a measurable contributor.
C — VEGAS Falsification	Does VEGAS fail on feature-degraded images?	$\Delta\text{CHAIR}_i = 0.0000$ all groups; $\alpha \approx 0$ for all images.	VEGAS cannot activate on this dataset; attention signal is uniformly diffuse.

3. **Track C confirms that VEGAS has no measurable effect.** Because $\alpha \approx 0$ universally (a direct consequence of the Track A finding), VEGAS's steering is disabled in practice on this dataset.

6.4 Limitations

1. **Sample size.** $n = 500$ images from a single split of COCO 2017. The Track B effect sizes ($\Delta\text{acc} \approx -0.035$) are statistically significant but modest; replication on a larger or different dataset is needed.
2. **Track C implementation.** Our VEGAS implementation approximates hidden-state steering. The near-zero α values likely reflect the fundamental uniformity of rollout attention (Track A finding) rather than an implementation artifact. Running VEGAS with fixed $\alpha > 0$ would test interpretation 2 above directly.
3. **Causality.** Track B is correlational. Track C was designed to convert the correlation to a causal claim, but the zero-α result means the causal test did not execute as intended. Alternative causal tests (patch ablation at layer 18, clean-image patch swap) remain to be run.
4. **Probe categories.** Only 17 of 80 COCO categories had sufficient positive examples (≥ 20). Rare categories may behave differently.

6.5 Contributions

This project makes the following empirical contributions:

1. **Track A null + Track B positive dissociation.** On the same 500-image dataset with the same model, attention geometry and feature content yield opposite results. To our knowledge this is the first paper to explicitly dissociate these two signals within the same VLM on the same hallucination ground truth.
2. **Layer-resolved probe analysis.** The dissociation holds at all four probed layers, with layer 18 as the sweet spot. This provides a operationally useful intervention target.
3. **Empirical grounding for the VEGAS blind spot.** The Track A and C results together provide data-backed evidence — not a conjecture — that VEGAS cannot address the failure mode Track B identifies.

7 Conclusion

We ran a controlled three-track empirical investigation into whether LLaVA-1.5-7B's ViT visual encoder is already failing on images that lead to hallucination.

The short answer: yes, but not in the way the literature expected.

Track A found no signal in attention. Track B found a consistent, significant signal in feature content. Track C found that the attention-based mitigation method VEGAS cannot activate its steering on this dataset because rollout attention is uniformly diffuse (a direct consequence of Track A's null), and therefore produces zero change in hallucination rates.

Together, the three tracks tell a coherent story: the ViT's failure on hallucinating images is a feature-content problem, not an attention-geometry problem. Methods that trust ViT attention as a corrective signal are addressing the wrong quantity. The practical implication is that future hallucination mitigation for LLaVA-style models should target patch embedding quality at mid-late ViT layers (around layer 18), not attention redistribution.

7 References

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A Track A: Additional Statistics

Table 7: Per-layer entropy gap: hallucinating vs. clean images. The largest gap occurs at layer 19 ($\Delta = 0.130$ nats), but is small relative to within-group variance.

Layer	Gap Δ (nats)	Direction
Layers 1–13	≈ 0.01–0.02	hallucinating > clean
Layer 14	≈ 0.00 (crossing)	—
Layers 15–24	increases to max 0.130 at layer 19	hallucinating > clean

B Track B: Probe Categories

The 17 COCO categories with ≥ 20 positive examples in the 500-image sample: *backpack, bed, bench, book, bottle, bowl, car, cell phone, chair, cup, dining table, handbag, laptop, person, sink, truck, umbrella.*

C Track C: Observed α Distribution

All 500 images had $\alpha \in [0.002, 0.010]$ with mean ≈ 0.005. This is consistent with the Track A finding that rollout entropy is nearly maximal ($\bar{H} \approx 6.321$ nats, maximum $= \log 576 \approx 6.356$ nats). The gap between observed mean entropy and maximum entropy is only ≈ 0.035 nats, which maps to $\alpha \approx 0.005$. VEGAS steers by $(1 + \alpha) \approx 1.005$ — effectively unity.